

1952



The hydrogen bomb test on the Enewetak Islands in the Pacific was the first thermonuclear explosion that combined atomic fusion and fission. Its mushroom cloud rose 10 miles (17 km) into the air.

THE YEAR 1952 saw some important breakthroughs. American microbiologist **Alfred Hershey** and geneticist **Martha Chase** worked together at New York's Cold Spring Harbor Laboratory to tackle a key question in biology: was **life's genetic material** made from protein or DNA? Some scientists

of highly specific wavelengths. Walsh developed the **atomic absorption spectrometer**, which measured this radiation and detected the tiniest levels of elements in a mixture. The spectrometer, patented the following year, later became a standard tool in forensic science and other fields that

12 MEGATONS OF TNT

THE SIZE OF THE 1952 IVY MIKE EXPLOSION

thought only protein was sufficiently complex to suit the task. Hershey and Chase examined the genetic material that phages—viruses that infect bacteria—inject into host cells. They discovered that the injected cells contained phosphorus—an element found in DNA, but not in protein. This indicated that genes were made of DNA.

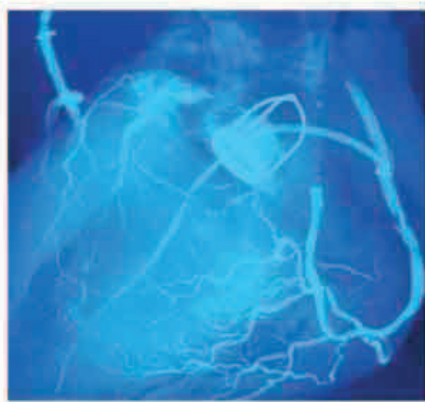
On the other side of the world, **Alan Walsh**, a British physicist living in Australia, pioneered a technique that revolutionized analytical chemistry. He explored the practical applications of the fact that atoms of different elements absorb radiation

demanding high-precision chemical analysis.

In the US, **Walton Lillehei** and **John Lewis** performed the first **open-heart surgery** in September. They induced hypothermia (cooling below normal body temperature) in their patient, giving them 10 minutes to

Artificial heart valve

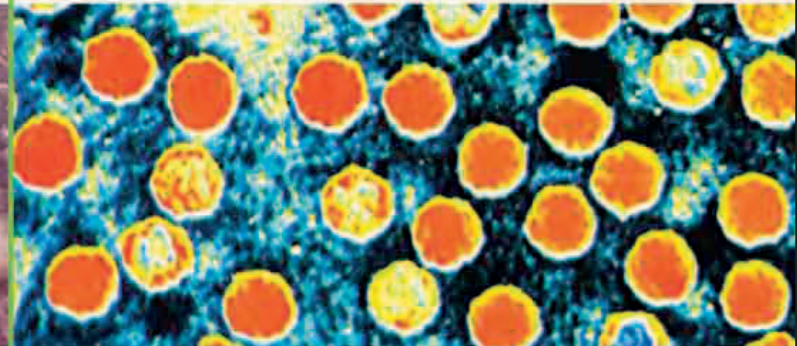
The first artificial heart valve was a caged-ball design. Blood leaves the heart as the ball is pushed against the cage. When the heart relaxes, the ball falls back to seal the valve.



correct a congenital heart defect. Just over a week later, American surgeon **Charles Hufnagel** implanted the first **artificial heart valve** in a patient with rheumatic fever, prolonging her life by nearly a decade. Plans were also made for the first **separation of conjoined twins**. The Brodie brothers, fused at the head, had been born the year before. **Oscar Sugar** and his team separated the twins and saved one of them.

On November 1, the first **hydrogen bomb** test—codenamed **Ivy Mike**—was conducted by the US on the Enewetak Islands in the northwest Pacific. It produced a 3 mile (5 km) fireball and obliterated a small island. Previous thermonuclear bombs had used atomic fission (see 1938), but Ivy Mike showed that an explosion could come, at least in part, from fusion (see 1988–89) too.

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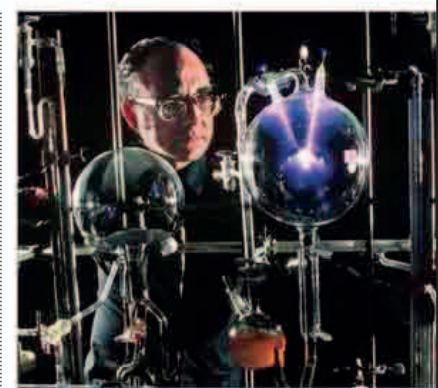
The polio vaccine, developed in the early 1950s against a virus that attacked the central nervous system, saved thousands from a lifetime of paralysis.

IN 1953, SCIENTISTS gained insight into how **inheritance and reproduction** worked at a **chemical level**. English scientists **James Watson** and **Francis Crick** (see pp.284–85) believed that DNA was the key. While the chemical makeup of DNA was partially known, the physical arrangement of its components had until then been a mystery.

By 1953, a new technique—**X-ray crystallography**—was being used to produce 3-D images of the structure of complicated biological molecules. At Kings College, London, a team that included Rosalind Franklin and Maurice Wilkins used it to study DNA. Franklin perfected a technique of preparing samples of DNA that yielded especially clear results. The early indications were



X-ray diffraction image of DNA
Franklin's remarkable, distinct X-ray diffraction image with its striking "X" pattern clearly indicated that DNA was shaped like a double helix.



that DNA was shaped like a helix. In spite of Franklin's reluctance to draw premature conclusions, Watson and Crick began building a **helical model of DNA** based on the evidence, and published their results in the science journal *Nature*. They depicted DNA as two molecular chains entwined around one another in a double helix. This model of DNA was ground-breaking, suggesting a way living things could reproduce their genetic material.

A month later, another journal, *Science*, reported on an experiment conducted the year before. American researchers **Stanley Miller** and **Harold Urey** had tried to re-create the **origin of life in a laboratory**. They had heated a mixture of ammonia, water, methane, and hydrogen in a sealed flask, and sparked it with electrodes to simulate lightning. Within two weeks the mixture was

